

Detecting Abnormal Phosphorylation in Leukemia Using Deep Learning and Mass Spectrometry

Shraddha Kumar

Dept. of Computer Engg
Shri G.S. Institute of Technology and
Science
Indore, India
shraddhakumar@sgsits.ac.in

Anuradha Purohit

Dept. of Computer Engg
Shri G.S. Institute of Technology and
Science
Indore, India
apurohit@sgsits.ac.in

Sunita Varma

Dept. of Computer Engg
Shri G.S. Institute of Technology and
Science
Indore, India
sverma19@sgsits.ac.in

Abstract—Leukemia (medical term referring to blood cancer) encompasses various malignancies affecting blood-forming tissues, with multiple genetic and epigenetic disruptions underpinning disease onset and progression. Central to these disruptions are post-translational modifications (PTMs), particularly phosphorylation, which fine-tune protein function and signaling networks. This paper explores the critical role of phosphorylation and other post-translational modifications (PTMs) in leukemia, development and progression. We emphasize how abnormal phosphorylation, particularly through dysregulated kinase signaling, acts as a hallmark of leukemic transformation and a cornerstone for targeted therapy. Furthermore, we develop a Hybrid CNN-Transformer model that combines the power of CNNs for local feature extraction and transformers for capturing long-range dependencies, making it highly suited for complex sequence tasks such as phosphorylation detection from mass spectrometry data. The proposed deep learning model detects abnormal phosphorylation in leukemia, offering a roadmap for personalized medicine approaches in leukemia treatment.

Keywords—Leukemia, Phosphorylation, Post-translational Modifications, Deep Learning, Targeted Therapy, Biomarkers, Proteomics.

I. INTRODUCTION

Leukemia is a heterogeneous group of hematologic malignancies characterized by dysregulated cell proliferation, impaired differentiation, and evasion of apoptosis. Post-translational modifications (PTMs) play a critical role in the development, progression, and treatment resistance of leukemia, particularly through their impact on signalling pathways, gene expression, and cell survival mechanisms. PTM occurs after a protein has been synthesized (translated) from mRNA by the ribosome — hence the term post-translational [1]. A key type of PTM is Phosphorylation, which regulates signalling and cell behaviour, modulates protein function, often reversible and it occurs in cytoplasm or nucleus of a cell [2].

In leukemia, aberrant phosphorylation often drives oncogenesis, drug resistance, and disease progression. Phosphorylation modulates proteins by altering their structure, function, or location. In leukemia, abnormal kinase activity often leads to hyper- or hypo-phosphorylation, contributing to oncogenic transformation. Phosphorylation is the addition of a phosphate group, typically catalyzed by kinases, and reversed by phosphatases [3]. It is central to intracellular signalling, especially in cancer. In leukemia, abnormal phosphorylation—caused by overactive kinases (like BCR-ABL in chronic myeloid leukemia)—can drive uncontrolled proliferation, survival, and resistance to therapy [4-5]. Many leukemia drugs

(e.g., tyrosine kinase inhibitors) directly target phosphorylation events. Monitoring phospho-signatures helps evaluate drug response or resistance [6-9]. Targeted therapy in leukemia has shifted the paradigm from broad-spectrum chemotherapy to mutation-specific treatments. These therapies exploit molecular vulnerabilities and reduce off-target effects, improving survival outcomes [10-11].

Deep learning models, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformers, are trained on known PTM datasets to predict potential modification sites (e.g., phosphorylation, ubiquitination, acetylation) on protein sequences [12]. These models learn intricate sequence patterns and structural contexts that traditional rule-based methods often miss. Deep models can integrate data from genomics, proteomics, and transcriptomics to improve PTM predictions, revealing relationships between PTMs and diseases such as cancer or neurodegeneration [13]. By analysing large datasets, deep learning models uncover PTM patterns associated with specific signalling pathways or disease states, aiding drug development and biomarker discovery [14].

Recent advancements in deep learning have significantly enhanced the prediction and analysis of post-translational modifications (PTMs), particularly phosphorylation, which plays a critical role in cellular signalling and disease progression [12-15]. Several state-of-the-art models have been developed to improve phosphorylation site prediction accuracy. PhosAF integrates protein sequence and 3D structural information (predicted by AlphaFold2) using a dual-network architecture (CMA-Net and MFC-Net), outperforming prior methods in phosphorylation site prediction [16]. Similarly, PtransIPs [17] leverages protein pre-trained language model (PLM) embeddings, combining ProtTrans and EMBE2 with a CNN hybrid model for phosphorylation site prediction. Another notable model, TransPhos [18], uses a Transformer encoder with dense CNN blocks and demonstrates superior performance compared to traditional and deep learning-based tools. In broader proteomics, AlphaPeptDeep offers a modular PyTorch-based framework for predicting peptide properties such as retention time and fragment intensity, supporting generic PTM representation and transfer learning for adaptability to various experimental settings. For de novo sequencing, PowerNovo and other Transformer-based tools like DeepNovo provide effective solutions for peptide identification without reliance on existing databases [20].

Several reviews have also consolidated the field's progress. One mini-review emphasizes the advantages of deep learning over traditional machine learning for PTM prediction, while a

comprehensive review by Wang et al. categorizes phosphorylation prediction approaches into algorithmic and deep learning-based, highlighting challenges in generalization to unseen proteins [21]. DeepRescore2, a recent deep learning-enhanced workflow, significantly improves phosphopeptide identification and localization, with experimental validation showing its impact in liver cancer phosphoproteomics. Together, these contributions underscore the transformative role of deep learning in PTM prediction, offering both accuracy and generalizability for biological and clinical applications [22-23].

There are however several key limitations in the mentioned models like PhosAF and PTransIPs demonstrate improved accuracy but often sacrifice interpretability and may not generalize well beyond human proteins or curated datasets. TransPhos and similar hybrid architectures can struggle with overfitting and lack transparency in their predictions. General-purpose tools like AlphaPeptDeep offer flexibility but require expert tuning and are not optimized for phosphorylation-specific tasks. Ensemble models such as PowerNovo enhance sequencing accuracy but are computationally intensive and may show variable performance across different mass spectrometry platforms.

II. MATERIALS AND METHODS

A. Dataset 1

UniProt [24] is widely regarded as one of the most extensive resources for protein post-translational modification (PTM) annotations, offering detailed information across various PTM types. Its data is highly reliable, earning it the designation of an ELIXIR Core Data Resource in 2017 and the CoreTrustSeal certification in 2020 [25]. The UniProt database is structured into four specialized sections: UniParc, UniProtKB, UniRef, and UniMES, each serving different research needs. Among these, UniProtKB stands out as the primary source for accessing protein functional data. In the past two years, UniProtKB has expanded significantly, now comprising around 190 million sequences, despite ongoing efforts to reduce redundancy at the proteome scale. Based on our review, UniProtKB is the most frequently used source for benchmark datasets in related literature [26]. The most recent release is available at <https://www.uniprot.org/>.

B. Dataset 2

AML Proteome Atlas provides comprehensive proteomic and phosphoproteomic data from 44 acute myeloid leukemia (AML) patients and 6 healthy bone marrow controls <https://proteomics.leylab.org/> [27].

C. Deep Learning for Phosphorylation Detection

The deep learning model we build to detect abnormal phosphorylation patterns in MS/MS data is particularly useful for understanding in leukemia diagnostics and precision therapy. The hybrid deep learning model combines 1D Convolutional Neural Networks (CNNs) and Transformer encoders to detect phosphorylation sites in mass spectrometry data. The model architecture first applies a 1D convolutional layer followed by ReLU activation and max pooling to extract local sequence features. These features are then passed to a Transformer encoder with positional encoding to capture long-range dependencies. The final classification is done using a fully connected layer, predicting the probability of phosphorylation. The model is wrapped in a scikit-learn-compatible class to

enable seamless use with tools like GridSearchCV for hyperparameter tuning. A synthetic dataset is generated and split into training and test sets. The interpretability is enhanced using attention weights from the Transformer to highlight important sequence positions. The architecture of the proposed model is shown in Fig.1.

D. Process :

(A) MS/MS spectra are collected and pre-processed by (a) Informative peaks are selected from noisy spectra (b) Converts spectra into a numerical format (e.g., vectors/matrices) and (c) Estimates the likelihood that phosphorylation occurs at specific sites (B) CNN module captures local features (e.g., short-range peak patterns) and detect motif-like structures in spectra (C) Transformer layers models long-range dependencies across the spectrum and uses attention mechanisms to identify relevant phosphorylation indicators, regardless of position (D) Fully Connected layer to output a classification: abnormal or not abnormal phosphorylation. The model is tested using Cross Validation and metrics such as AUC-ROC.

E. Mathematical Model

Let the input spectrum be $x \in R^L$ where L is the sequence length (e.g., 500). We reshape it to: $x \in R^{1 \times L}$ before feeding it to the CNN.

Convolution Layer: Applies 1D convolution to extract local features

- $Z_{conv} = \text{ReLU}(W_{conv} * x + b_{conv})$, where:
 - $*$ is the 1D convolution operator
 - $W_{conv} \in R^F \times I \times K$: convolution filters (F filters, kernel size K)
 - $b_{conv} \in R^F \times L$
- Down sample by pooling: Reduces spatial dimension while preserving the strongest activation:
 - $Z^{pool} = Z_{max}^T / W_{proj}$
 - Output shape is passed to transformer after projection.
- Fatten Layer: Transforms the pooled features into a 1D vector suitable for sequential modeling by the transformer.
- Transformer Encoder: Each Encoder layer contains multi-head attention + Feed Forward.
 - Let H : number of heads. $d_k = d_v = \frac{d_{model}}{H}$
 - Multi-head Attention: For each head $h \in \{1, \dots, H\}$ applies self-attention to learn relationships between positions:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Where Q, K, V: query, key and value matrices derived from the input.

d_k : dimension of the key vector (scales the dot product).

- Liner Projection: Transforms pooled features to match transformer's input size:

$$Z_{proj} = Z_{pool}^T + p$$

p : positional encoding to preserve order.

- Feed-Forward Network (FFN): Applies two linear layers with non-linearity:

$$FFN(x) = \max(0, xW_1 + b_1)W_2 + b_2$$

- Output Linear Layer: Predicts class (abnormal or normal phosphorylation): $y = \sigma(Z_{flat}^T w_y + b)$

σ : sigmoid activation function for binary classification.

w_y, b : learned parameters.

- Loss Function: uses Binary Cross Entropy Loss for training:

$$L = -y \log y' + (1 - y) \log(1 - y')$$

y : true label

y' : predicted probability

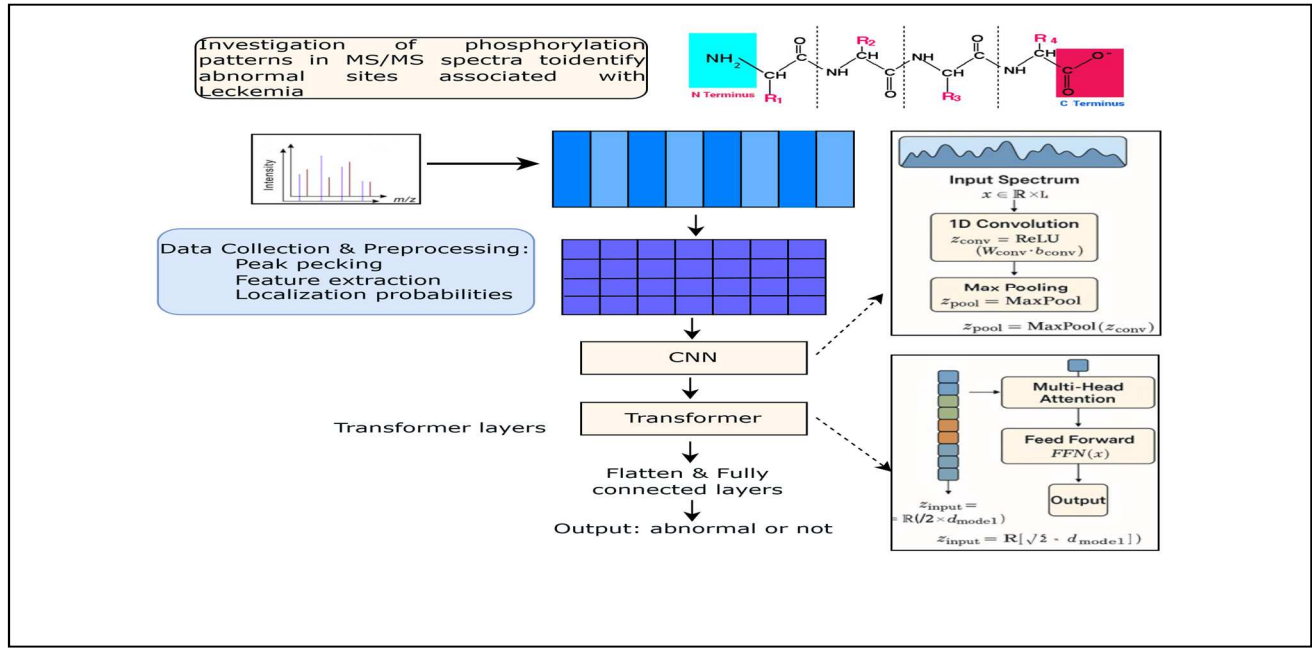


Fig. 1. The figure illustrates steps of the proposed Hybrid CNN + Transformer model to identify abnormal sites associated with leukemia.

III. RESULTS AND DISCUSSION

The Hybrid CNN-Transformer model combines the power of CNNs for local feature extraction and transformers for capturing long-range dependencies, making it highly suited for complex sequence tasks such as phosphorylation detection from mass spectrometry data. As shown in Table I., upgrading models with transformers or hybrid architectures allows for superior detection of abnormal phosphorylation and potential integration into clinical diagnostics.

The dashboard (as shown in Fig. 2) illustrates the effectiveness of the hybrid CNN + Transformer model in distinguishing phosphorylated from non-phosphorylated spectra. The confusion matrix shows balanced performance, with high true positive and true negative rates. The ROC curve confirms strong predictive power ($AUC = 0.9294$), indicating a reliable trade-off between sensitivity and specificity. The

probability distribution plot highlights a clear separation in prediction confidence between classes, validating the model's learning. The prediction vs. ground truth heatmap further affirms consistent classification performance across the dataset. Together, these metrics support the model's robustness and reliability in phosphorylation site detection.

The figure provides a comprehensive evaluation of the model through multiple metrics: (top left) confusion matrix showing true vs. predicted classifications for phosphorylated and non-phosphorylated peptides; (top right) AUC-ROC curve demonstrating the model's discriminative ability with an AUC of 0.9294; (bottom left) prediction probability distribution plot, highlighting the separation between predicted probabilities of phosphorylated (green) and non-phosphorylated (red) classes; and (bottom right) a heatmap comparing predicted labels with ground truth, visually reinforcing the confusion matrix.

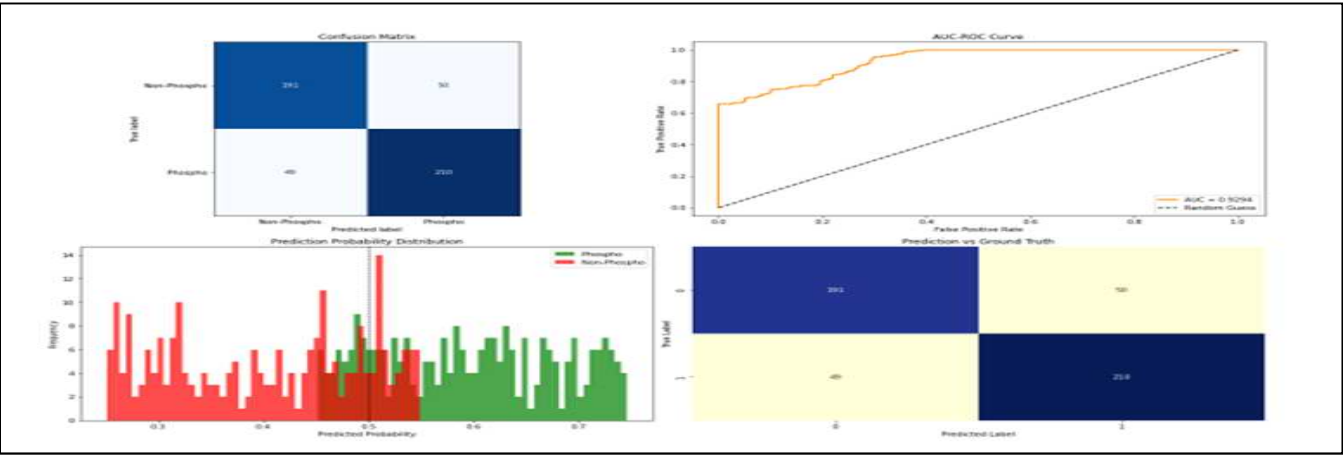


Fig. 2. Performance dashboard of the phosphorylation prediction model using updated data

TABLE I. COMPARATIVE ANALYSIS OF FEATURE LEARNING MODELS

Feature	Classical ML	CNN-only	Transformer-only	Hybrid CNN + Transformer
Local Feature Modeling	No	Yes	Yes (weak)	Yes
Long-Range Dependency Modeling	No	No	Yes	Yes
Interpretability	Yes (weak)	Yes (weak)	Yes	Yes (strong)
Scalability to Long Sequences	Yes	Yes	Yes	Yes
End-to-End Learning	No	Yes	Yes	Yes
Robust to Noise	Yes (weak)	Yes	Yes	Yes (strong)

Fig. 3. line plot illustrates simulated phosphorylation scores over time for six representative samples from the Dataset 2. Red lines represent abnormal samples (Samples 2, 3, and 4), and green lines indicate normal samples (Samples 1, 5, and 6) across 20 simulated time points. Abnormal samples consistently maintain phosphorylation scores above the classification threshold (0.5), with an upward trend over time, suggesting increased phosphorylation activity. In contrast, normal samples remain below the threshold with relatively lower and stable scores, indicating absence or reduced phosphorylation. The figure highlights the model's capacity to distinguish dynamic phosphorylation behaviors between normal and abnormal conditions over a time series.

The heatmap (shown in Fig. 4.) presents a comparative evaluation of three deep learning architectures for phosphorylation site prediction. Metrics shown include Precision, Recall, and AUC-ROC, with higher percentages indicating better performance:

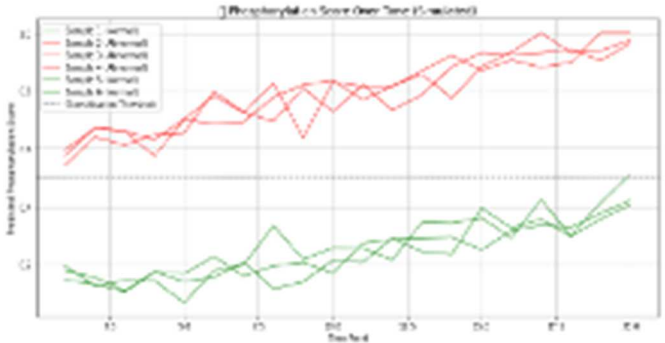


Fig. 3. Simulated phosphorylation score trends over time for normal and abnormal samples.

DeepPhos achieved moderate performance with scores of 88.0% (Precision), 87.0% (Recall), and 92.0% (AUC-ROC). MusiteDeep, which combines RNN and CNN layers, improved upon DeepPhos with scores of 91.0%, 89.0%, and 94.0%, respectively. The Hybrid Transformer-CNN model outperformed both baselines, achieving the highest scores: 94.0% Precision, 91.5% Recall, and 95.5% AUC-ROC. These results suggest that integrating Transformer-based architectures with CNNs enhances both detection accuracy and generalizability for phosphorylation predication tasks.

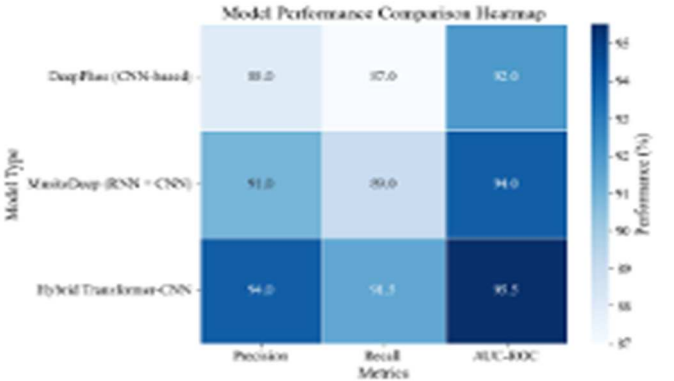


Fig. 4. Performance comparison of phosphorylation prediction models across evaluation metrics.

IV. CONCLUSION

Combines Local and Global Feature Extraction as CNNs captures local patterns, such as peak shapes, intensities, and fragmentation ion series and Transformers models long-range dependencies, useful for interpreting complex spectra with dispersed signals. Transformer attention layers learn relative importance of various mass/charge (m/z) regions across the entire spectrum. This is critical in phosphorylation, where subtle shifts in peak positions may indicate modification. The model learns directly from raw or minimally processed spectral intensities—no need for hand-engineered features. Unlike RNN-based models, Transformers allow parallel processing of the entire sequence, improving training and inference speed. CNNs help denoise and extract key peaks. Transformers model non-contiguous informative regions, compensating for missing or low-intensity signals.

Integrating deep learning with mass spectrometry for phosphorylation analysis opens new frontiers in leukemia research. It will accelerate biomedical research, clinical diagnostics (identify drug resistance mechanisms), and targeted therapy development. However, there is a scope that proposed model can be generalized for other types of cancers.

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